SEINARTS AUDIT REMEDIATION

Remediation Ledger Reader's Edition

A human-readable narrative and checklist rendering of the canonical engineering ledger

14	3	2	42
FORMALLY CLOSED	IN PROGRESS	APPROVED	OTHER OPEN

61 tracked rows. Formal closure is 14 of 61 (22.95%). This is a strict outcome count, not an effort-weighted completion estimate. The canonical source remains Docs/Engineering/AuditRemediation.md.

Presentation snapshot: July 29, 2026

Source branch: codex/audit-remediation

Purpose: planning, review, and human status inspection

Contents

SeinARTS Audit Remediation

Reader's edition

This document is a presentation-oriented rendering of the canonical AuditRemediation.md. It preserves the findings, phases, statuses, approved decisions, evidence, and completion gates recorded in that ledger while replacing the dense tracking tables with narrative sections and checklists.

The engineering ledger remains the canonical audit trail. This copy reflects its working-tree state on July 29, 2026; it is not a replacement for it and its completion counts are not effort-weighted.

Executive status

The campaign tracks **61 findings and approved work items**.

- **14 are formally closed:** 13 Verified and 1 Fixed.
- **3 are In progress.**
- **2 are Approved for implementation but are not complete.**
- **26 are Confirmed and awaiting their implementation or verification wave.**
- **7 are Queued for phase-local revalidation.**
- **9 are Gates requiring a product or API decision.**
- Nothing is currently marked Disproved or Deferred.

That is **14 of 61 rows, or 22.95%, formally closed** under the ledger's strict Definition of Complete. It is deliberately not an estimate of engineering effort completed: foundational work can unlock several rows, while a single acceptance row can require substantial multi-process or PIE evidence.

By workstream:

- **Correctness, determinism, and lifecycle:** 27 items; 13 formally closed.
- **Performance and memory:** 11 items; none formally closed.
- **API, modularity, and extensibility:** 14 items; 1 formally closed.
- **Feature and completeness scope:** 9 items; none formally closed.

How to read the checklists

- `[x]` means the row is formally closed as **Verified** or **Fixed**.
- `[ ]` means the row remains open, including **In progress**, **Approved**, **Confirmed**,

Queued, and **Gate** states.

- The explicit status beside every item is authoritative. **Approved** means the direction is

approved, not that implementation is complete. **Fixed** and **Verified** remain distinct: Verified includes the required broader build, determinism, or integration gates.

Status vocabulary:

- **Queued:** the audit finding awaits phase-local revalidation.
- **Confirmed:** the finding was reproduced or re-grounded against the branch-point code.
- **In progress:** implementation or verification is active.

- **Fixed:** implementation and focused regression coverage are complete.
- **Verified:** required build, determinism, and integration gates are also green.
- **Disproved:** live behavior or a test refuted the audit claim and the evidence is recorded.
- **Gate:** RJ must make the product or API decision before implementation.
- **Deferred:** RJ explicitly accepted deferral with a rationale.

Baseline and evidence timeline

Starting point

The campaign started from branch point `d22ba7101ea74a88d88dbf99294ab391384535d5` on `codex/audit-remediation`. The baseline `SeinARTSEditor Win64 Development` build succeeded on July 18, 2026 in 25.77 seconds.

Production source at baseline, counting `.h`, `.cpp`, and `.cs` files:

- **Host Source:** 5 files, 65 lines, 16 blank lines, 10 comment-like lines.
- **Framework:** 508 files, 96,707 lines, 11,943 blank lines, 29,192 comment-like lines.
- **Squad extension:** 18 files, 2,018 lines, 228 blank lines, 527 comment-like lines.
- **Cover extension:** 33 files, 4,083 lines, 431 blank lines, 1,386 comment-like lines.
- **Movement+ extension:** 18 files, 2,361 lines, 283 blank lines, 828 comment-like lines.

These are review metrics, not reduction quotas. Production code, tests, and comments are tracked separately.

Phase 0 - test infrastructure

- The disabled base `Plugins/SeinARTSTestSuite` and disabled

`Plugins/SeinARTSExtensionTestSuite` separate tests from production and deny all test modules in Shipping.

- The extension suite has separate DeveloperTool and Editor modules, permitting a true

framework-only build and run profile without mounting extension assets.

- Production modules depend on neither the test plugins nor `CQTest`.
- The test-enabled Editor build succeeded on July 18, 2026.
- A headless `SeinARTS.Unit` smoke test was discovered and passed. The runner parsed the exported

`index.json` instead of trusting only the process exit code.

- Both `All` and extension-disabled `Framework` profiles built and executed. The Framework receipt

marked Squad, Cover, Movement+, and the extension test suite disabled.

- The strict runner rejected seven pre-Automation asset-load errors that Unreal otherwise reported

as green. `-AllowKnownStartupErrors` accepts only the exact sorted checked-in signature multiset while `CONTENT-01` remains tracked.

- Test-profile builds restored the ordinary Editor receipt before Automation launched, so a test

run does not leave test plugins enabled or production extensions suppressed.

- The ordinary Editor build after scaffolding contained no Sein test plugin or module entries.
- The baseline Shipping build correctly excluded tests but failed on an editor-only

`UTexture2D::MipGenSettings` reference in FoW rendering. That became `BUILD-01`.

Phase 1 - fixed-point content migration

An all-content `LoadPackage -all -projectonly` scan found exactly seven packages retaining the pre-native `FFixedPoint` tagged layout. A temporary read-only compatibility path recovered each raw 32.32 value, the packages were resaved individually, and the compatibility code was removed. A native full-content reload then produced zero tagged-property mismatches and reproduced this exact manifest:

- `SA_Build`: `1,288,490,188,800`
- `SA_Place_Barracks`: `214,748,364,800`
- `SA_Place_VehicleDepot`: `214,748,364,800`
- `SA_Produce_CombatCar`: `64,424,509,440`
- `SA_Produce_Research_VehicleDepot`: `107,374,182,400`
- `SA_ThrowSmoke`: `2,147,483,648,000`
- `SE_UnlockVehicleDepot`: `-4,294,967,296`

The same scan found four unrelated obsolete root assets with broken redirected imports: `BP_Unit`, `SU_Basic`, `SU_Wheeled`, and `LVL_Lobby`. They remain isolated as `CONTENT-02` rather than being mixed into the serializer migration.

Phase 1 - correctness verification

- The full `All` profile passed 74 of 74 tests. Focused evidence included Effects 30/30, Snapshot

4/4, Determinism 3/3, placement/cover integration 2/2, and editor-style lifecycle 1/1. The Framework-only unit profile passed 68/68 before the final all-category run.

- Effect tests covered globally unique IDs, source-aware ability ownership, callback-time

removal/destruction, stack/tag saturation, transactional rollback, and deterministic iteration. An adversarial concern about nested away/back transfer losing one of multiple detached ability classes was disproved by an exact two-class lifecycle test.

- Snapshot v4 rejected dirty free slots, abstract/deprecated/newer classes, malformed or trailing

tagged bytes, cross-wired ability owners, and contradictory primary/passive roles before mutating live state. A custom `Within=SeinWorldSubsystem` implementation also round-tripped.

- A same-process serial/parallel collision workload passed. Two fresh Editor processes ran the same

100-collider workload for 120 production ticks: all 120 raw fixed-point pose digests matched while all 120 then-existing `StateHash` values differed. This isolated `STATE-02`; fresh-process StateHash agreement was **not** claimed.

- Clean MSVC Development and Shipping builds succeeded. Their receipts excluded Sein test plugins

and products as required.

- Clang compiled and linked `SeinARTSCore` and `SeinARTSCoreEntity` after fixed-point division was

moved off compiler-emitted 128-bit runtime helpers. A full Clang Editor executable could not link against Epic's MSVC launcher binaries, so Clang execution was **not** claimed.

- `CompileAllBlueprints` found zero Blueprint compile errors. Seven warnings remained: two Python

name collisions (`ExtentsShape`, `StampShape`), two missing BrandKit `SeinAssetIcon92.png` references, and three invalid gameplay tags (`SeinARTS.Ability.Garrison`, `SeinARTS.Unit.Vehicle`, and `SeinARTS.Tech.VehicleDepot`). The native all-content load had zero serializer mismatches and only the four separately tracked `CONTENT-02` failures.

Phase 1 added 2,534 production lines overall. That increase documents correctness and explicit network/snapshot contracts; it is evidence, not a target. Production counts at that checkpoint:

- **Host Source:** 5 files, 65 lines, delta 0, 16 blank, 10 comment-like.
- **Framework:** 508 files, 99,222 lines, delta +2,515, 12,150 blank, 29,255 comment-like.

- **Squad extension:** 18 files, 2,026 lines, delta +8, 229 blank, 526 comment-like.
- **Cover extension:** 33 files, 4,094 lines, delta +11, 432 blank, 1,377 comment-like.
- **Movement+ extension:** 18 files, 2,361 lines, delta 0, 283 blank, 828 comment-like.

Phase 2 - command protocol and replay foundation

- The test-enabled `All` profile passed 172/172 `SeinARTS.Unit` tests. The exported report is

`Saved/Automation/SeinARTS.Unit-20260722-122403/index.json`: 153 clean passes, 19 expected-warning passes, no failures, and no skipped tests.

- Raw `FName` command payloads moved to a frozen manifest-bound catalog with bounded `uint32` wire

indices. Hostile bytes cannot intern names, and standalone, network, and replay paths normalize to the same catalog representative.

- The recorded compatibility boundary was command wire v3, command protocol domain 6, replay file

v6, and header metadata v5.

- Executable replay and header-metadata serialization use the active world's immutable catalog.

Literal-null metadata calls retain an explicit worldless-tooling fallback; a supplied context without a frozen simulation world fails closed.

- Adversarial review exposed the tick-zero bootstrap barrier and shared materialization problem

despite the green suite. It remained visible as `STATE-03`.

Phases 3 and 4 - tick zero, authority, and lifecycle

Gate G was implemented as a world-lifetime inert facade and one transient bootstrap transaction: freeze contract, materialize, seal and preflight, compare exact local receipts, authorize every simulation participant, release the transaction, and launch the already-reserved dormant tick-zero scheduler. Failure is terminal for that world; reconnect and checkpoint adoption do not reopen bootstrap.

Additional evidence and contracts:

- `ASeinGameMode` remains Unreal's match authority/controller shell without owning parallel

deterministic defaults. Legacy default-faction and GameMode starting-resource paths were removed. Active Human/AI slots require explicit valid factions. Framework starting resources are a canonical match extension. The default materializer requires exactly one `ASeinPlayerStart` per active Human/AI slot; custom materializers remain pluggable.

- The canonical receipt binds the frozen contract, authorization context, materialization plan, and

initial state. Native contributors use registration handles with teardown. Blueprint and extension code can contribute immediate deterministic `FInstancedStruct` values. Seed installation and Applying-phase mutators require the exact lexical materializer capability. Legacy mutable storage access remains tracked as `API-14`.

- Bootstrap consensus is transport-neutral. The shipped Unreal relay supports standalone,

dedicated, and listen-server topologies without encoding those shapes into authority policy. Pending travel state is bound to source world and destination, same-map travel is explicit, failure delegates are removed during deinitialization, and irreversible launch occurs only after unanimous authorized-ready.

- Command ingress remains closed until `Consumed + running`. Local drafts, authenticated transport,

replay, deterministic-system, AI, and observer routes have distinct capabilities and lifecycle gates. AI emission is exact-controller and lexical; replay primes commands at a private exact-tick boundary; public completion/bootstrap callbacks run in the guarded read-only observer scope.

- State hardening includes live-only guarded entity/component reads, exact deferred-destroy inspection, generation-preserving component iteration, generation-exhaustion retirement, staged snapshot-v7 restore, snapshot reentrancy/GC protection, and explicit ownership-transition quiescence. Full continuation snapshots and canonical cross-process hashing remained Phase 5.

- Independent adversarial review found no source-level Phase-4 blocker. The `All` profile passed Unit

210/210, Sim 27/27, Determinism 5/5, Integration 10/10, and Editor 1/1. The extension-disabled Framework profile passed Unit 210/210, Sim 26/26, Determinism 5/5, and Integration 9/9. Representative reports are `Saved/Automation/SeinARTS.Unit-20260723-000315`, `SeinARTS.Sim-20260723-000947`, `SeinARTS.Determinism-20260723-001010`, `SeinARTS.Integration-20260723-001409`, and `SeinARTS.Unit-20260723-001507`.

- Ordinary Development and Shipping builds succeeded. Neither receipt enabled a Sein test plugin or

contained a Sein test build product, and production source had no `CQTest` or test-suite dependency.

- A real multi-peer travel/launch `SeinARTS.Network.PIE` scenario was still absent, so `STATE-03`

remained **Fixed** instead of **Verified**.

Phase 4 - source-size checkpoint

The Phase-4 correctness work did **not** shrink production source. Relative to Phase 1, Framework production grew by 23,631 lines and 40 files. Most of the growth is executable protocol, validation, serialization, and lifecycle machinery rather than comments, but it is not an acceptable final readability posture: `SeinWorldSubsystem.cpp` reached 9,615 lines and `SeinNetSubsystem.cpp` reached 6,401. `API-13` tracks internal decomposition after the exact-state contracts are frozen. Reduction must preserve tested seams and execution order.

Counts at that checkpoint:

- **Host Source:** 5 files, 65 lines; delta from baseline 0, delta from Phase 1 0; 16 blank and 10

comment-like.

- **Framework:** 548 files, 122,853 lines; delta from baseline +26,146, delta from Phase 1 +23,631;

13,737 blank and 30,219 comment-like.

- **Squad extension:** 18 files, 2,056 lines; delta from baseline +38, delta from Phase 1 +30; 231

blank and 531 comment-like.

- **Cover extension:** 33 files, 4,093 lines; delta from baseline +10, delta from Phase 1 -1; 432

blank and 1,376 comment-like.

- **Movement+ extension:** 18 files, 2,361 lines; delta from baseline 0, delta from Phase 1 0; 283

blank and 828 comment-like.

Approved architectural and product decisions

- Effects use one simulation-global deterministic identity namespace.
- Authored destinations may overrule coarse static-nav false negatives and their owning provider's

obstruction, but not unrelated live blockers, occupants, reservations, or hazards.

- Tactical cover matching and reservations are implementation scope, not an idea backlog.
- Legacy FoW cell output is not a compatibility requirement. A faster deterministic default wins

when it meets or improves truthfulness, resolution, responsiveness, shapes, layers, and visual quality.

- Crisp contracts are automated; movement, tactical, and presentation feel remains an RJ PIE

decision.

- Tests stay in clearly separated non-shipping organization. Production modules do not acquire

`CQTest` dependencies merely to host a suite.

- Touched production code should become smaller and clearer when that follows naturally from the

fix. Raw line reduction never outranks correctness.

- Plugin-local `AGENTS.md` guides are canonical for Codex. All `CLAUDE.md` files remain for possible

future Claude use.

- Work remains in the main checkout; worktrees are forbidden.
- Command authority is topology-neutral. Transport-authenticated participants, gameplay player

slots, simulation/hash peers, coordinator capability, and match-administration capability are separate identities and roles. Coordinating a turn does not grant gameplay or administration rights.

- The default authority policy is owner control plus explicit deterministic scoped grants. Broker

orders retain mixed-recipient behavior by filtering unauthorized or stale members; single-entity commands reject unauthorized control. Delegated actions spend the entity owner's resources by default, while custom policies may implement shared-resource games.

- Commands use an exact registered tag and schema-version contract with bounded deterministic

payloads. Native built-ins and Blueprint/extension handlers share the frozen compatibility manifest. Missing or conflicting handlers fail before tick zero instead of being silently skipped.

- The framework ships the existing Unreal dedicated/listen relay as its first transport adapter,

while authority and turn aggregation remain compatible with future elected-P2P and all-peer adapters. Authenticated fail-stop peers are the P2P baseline; Byzantine consensus and anti-DoS are outside the framework guarantee.

Correctness, determinism, and lifecycle checklist

[x] **COR-01 - Verified**

- **Phase:** 2/4
- **Finding:** Multiplayer command ownership, sender-role, match-control, payload, and turn-window validation are incomplete.

[x] **COR-02 - Verified**

- **Phase:** 1
- **Finding:** Effect IDs collide across scope/storage; removal identity is ambiguous.

[ ] **STATE-01 - Confirmed**

- **Phase:** 5
- **Finding:** Snapshot capture/restore is not exact continuation state across all future-affecting systems, including centralized ability active-index lifecycle.

[x] **COR-03 - Verified**

- **Phase:** 1
- **Finding:** Latent-action iteration can be invalidated by synchronous Blueprint callbacks.

[ ] **STATE-02 - Confirmed**

- **Phase:** 5
- **Finding:** StateHash coverage/canonicalization is incomplete and includes process-local `FName` identity.

[x] **STATE-03 - Fixed**

- **Phase:** 3/4
- **Finding:** Tick-zero bootstrap lacked a canonical completion barrier and replay-compatible shared materialization contract; server and client derived GameMode defaults and failure paths differently.

[ ] **NAV-01 - Confirmed**

- **Phase:** 5
- **Finding:** Initial destinations can be silently moved after preview by partial A*, wall push, authority recognition, or endpoint restoration.

[ ] **NAV-02 - Confirmed**

- **Phase:** 5
- **Finding:** Budgeted asynchronous repath results can be overwritten or lost when keyed only by requester.

[ ] **CACHE-01 - In progress**

- **Phase:** 1/6
- **Finding:** Structured-XOR and fingerprint keys can collide; FoW source/blocker rotation/invalidation is incomplete. Exact nav/FoW cache identities and equal-cell reset are fixed; broader invalidation remains.

[ ] **NET-01 - In progress**

- **Phase:** 1/4/5/8
- **Finding:** Failed local submission, map restart, retained hash/turn sets, and pruning lifecycle were incomplete. Readiness, ownership, exact travel binding, failure cleanup, retry, bounds, slot/turn, and reset contracts are fixed; checkpoint catch-up and long-session retention remain.

[x] **COR-04 - Verified**

- **Phase:** 1
- **Finding:** Free-rotation placement validates an incorrect or default yaw.

[x] **COR-05 - Verified**

- **Phase:** 1
- **Finding:** Collision overlap pair identity omits entity generation.

[ ] **NAV-03 - Queued**

- **Phase:** 5
- **Finding:** Same-cell vehicle paths can contain no drivable segment and fail incorrectly.

[ ] **MOVE-01 - Gate**

- **Phase:** 7
- **Finding:** Fixed-wing idle/coast behavior can violate continuous-flight expectations.

[x] **COR-06 - Verified**

- **Phase:** 1
- **Finding:** Generic component initialization can run twice.

[x] **EDIT-01 - Verified**

- **Phase:** 1
- **Finding:** Editor-owned rooted resources are not reliably released.

[x] **COR-07 - Verified**

- **Phase:** 1
- **Finding:** Player-effect callbacks may depend on unordered storage iteration.

[ ] **FOW-01 - Queued**

- **Phase:** 6
- **Finding:** No-bake/runtime fallback crosses nondeterministic float/editor behavior into authoritative visibility.

[ ] **FOW-02 - Confirmed**

- **Phase:** 6
- **Finding:** FoW stores one blocker height beside an OR'd layer mask, so overlapping layers incorrectly inherit the tallest blocker.

[ ] **FOW-03 - Confirmed**

- **Phase:** 6
- **Finding:** Dynamic FoW blocker height ignores the authored extents `LocalOffset.Z`.

[ ] **FOW-04 - Confirmed**

- **Phase:** 6
- **Finding:** Terrain vision scaling adjusts radial and rectangular shapes but omits cone length.

[x] **MATH-01 - Verified**

- **Phase:** 1
- **Finding:** Extreme fixed-point construction and arithmetic contain signed-overflow and undefined-behavior edges.

[x] **CFG-01 - Verified**

- **Phase:** 1
- **Finding:** Cover's simulation-affecting settings are not yet a stable config-fingerprint contributor.

[x] **BUILD-01 - Verified**

- **Phase:** 1
- **Finding:** Shipping compile references editor-only `UTexture2D::MipGenSettings` in FoW rendering.

[ ] **BUILD-02 - Confirmed**

- **Phase:** 7
- **Finding:** Uncooked headless `-game` loads `SW_UnitBanner`, whose generated class depends on editor-only `USeinWidgetBlueprint`; the asset class needs an UncookedOnly/runtime-safe ownership seam.

[x] **CONTENT-01 - Verified**

- **Phase:** 1/7
- **Finding:** Existing Blueprint assets emit tagged-property deserialization errors during headless startup.

[ ] **CONTENT-02 - Confirmed**

- **Phase:** 7
- **Finding:** Four obsolete root-level assets have broken redirected imports during an all-content load.

Performance and memory checklist

- [ ] **PERF-01 - Confirmed**
  - **Phase:** 5/8
  - **Finding:** Redundant full StateHash walks occur at incompatible cadences.
- [ ] **PERF-02 - Confirmed**
  - **Phase:** 5/8
  - **Finding:** A* scratch allocation churn is high; retained worker contexts need an explicit memory cap.
- [ ] **PERF-03 - Confirmed**
  - **Phase:** 6
  - **Finding:** FoW changed-source footprint generation is roughly cubic in radius.
- [ ] **PERF-04 - Confirmed**
  - **Phase:** 6
  - **Finding:** Any dynamic FoW blocker change invalidates all sources rather than only spatially affected sources.
- [ ] **PERF-05 - Confirmed**
  - **Phase:** 6
  - **Finding:** Minimap performs dense point queries, buffer churn, and full texture recreation and upload.
- [ ] **PERF-06 - Confirmed**
  - **Phase:** 5/8
  - **Finding:** Cover provider and slot work has quadratic paths and duplicated allocation bodies.
- [ ] **PERF-07 - Confirmed**
  - **Phase:** 7/8
  - **Finding:** Squad, avoidance, and collision scan broad entity sets and allocate avoidable per-tick containers.
- [ ] **PERF-08 - Confirmed**
  - **Phase:** 5/8
  - **Finding:** Replay and completed/hash/turn histories can grow without practical bounds.
- [ ] **PERF-09 - Confirmed**
  - **Phase:** 8
  - **Finding:** High effect stack counts materialize one resolved modifier copy per stack.
- [ ] **PERF-10 - Confirmed**
  - **Phase:** 5/8
  - **Finding:** Merely enabling Cover binds the authority resolver and globally forces collision serial, even in worlds with no authoritative cover destination.
- [ ] **PERF-11 - Confirmed**
  - **Phase:** 5/8
  - **Finding:** StateHash parallel dispatch is budgeted by storage count, so the default minimum batch normally leaves the expensive entity walk serial.

API, modularity, and extensibility checklist

- [x] **API-01 - Verified**
    - **Phase:** 2/4
    - **Finding:** Command dispatch and validation are centralized as hardcoded branching rather than stable registered handlers and policies.
- [ ] **API-02 - Queued**
    - **Phase:** 1/7
    - **Finding:** Some public headers rely on dependencies declared private in `Build.cs`.
- [ ] **API-03 - Confirmed**
    - **Phase:** 7
    - **Finding:** Cover's optional Squad dependency is metadata-only; the bridge module hard-links Squad.
- [ ] **API-04 - Gate**
    - **Phase:** 7
    - **Finding:** Terrain query ownership leaks through Navigation rather than a neutral terrain contract.
- [ ] **API-05 - Confirmed**
    - **Phase:** 5/7
    - **Finding:** Single-cast delegates prevent deterministic composition of multiple providers.
- [ ] **API-06 - Queued**
    - **Phase:** 3/7
    - **Finding:** LevelData channel/schema identity and compatibility versioning are weak.
- [ ] **API-07 - Gate**
    - **Phase:** 5/7
    - **Finding:** Typed path kinds lack an extensible custom payload and validation story.
- [ ] **API-08 - Queued**
    - **Phase:** 5
    - **Finding:** Per-unit terrain restrictions are not wired through the full request and planner path.
- [ ] **API-09 - Queued**
    - **Phase:** 6/7
    - **Finding:** FoW lifecycle hooks and implementation contracts are advertised more broadly than wired.
- [ ] **API-10 - Gate**
    - **Phase:** 7
    - **Finding:** Targeter gestures and attribute modifier types are narrower than their Blueprint-facing promise.
- [ ] **API-11 - Confirmed**
    - **Phase:** 5
    - **Finding:** Cover authority is a single-cast Boolean with no requester, source, stable slot, or override-policy context.
- [ ] **API-12 - Confirmed**
    - **Phase:** 3/7

- **Finding:** Direct ability activate/end/cancel paths do not centrally maintain `ActiveAbilityID` and `ActivePassiveIDs`; snapshot restore preserves IDs but deliberately deactivates opaque passive execution.

[ ] **API-13 - Confirmed**

- **Phase:** 8
- **Finding:** Core and Net state machines are over-concentrated in 9,615-line and 6,401-line implementation files; split internal responsibilities without widening public seams or changing execution order.

[ ] **API-14 - Confirmed**

- **Phase:** 5
- **Finding:** Public mutable entity-pool, component-storage, and player-state accessors bypass the guarded mutation/bootstrap facade and can violate lifecycle invariants.

Feature and completeness checklist

- [ ] **FEAT-01 - Queued**
  - **Phase:** 5
  - **Work:** Checkpoint plus command-tail reconnect and exact catch-up.
- [ ] **FEAT-02 - In progress**
  - **Phase:** 5
  - **Work:** Replay journaling, checkpoints, seeking, validation, and bounded storage.
- [ ] **FEAT-03 - Approved**
  - **Phase:** 5
  - **Work:** Tactical cover matching, stable slot identities, reservations, lifecycle, and shared preview/commit planning.
- [ ] **FEAT-04 - Approved**
  - **Phase:** 6
  - **Work:** Faster height-aware FoW default plus spatial invalidation and performance/quality A/B.
- [ ] **FEAT-05 - Gate**
  - **Phase:** 7
  - **Work:** Rich targeter and gesture registry plus public policy composition.
- [ ] **FEAT-06 - Gate**
  - **Phase:** 7
  - **Work:** Team and shared FoW policy.
- [ ] **FEAT-07 - Gate**
  - **Phase:** 7
  - **Work:** Production, voting, and reinforcement completeness with stable identities.
- [ ] **FEAT-08 - Gate**
  - **Phase:** 7
  - **Work:** Vehicle typed-path producer and validation plus flight/3D behavior, consistent with steering-first movement and offline-authored curves.
- [ ] **FEAT-09 - Gate**
  - **Phase:** 7/8
  - **Work:** Adaptive input delay after observability and policy review.

Replay foundation boundary

The replay foundation recorded in the ledger is a bounded v6 executable format owned exclusively by `SeinARTSNet`. Full recordings start at tick zero, retain every applied assembled turn including empty heartbeats, stop on the exact inclusive `EndTick`, bind executable command decoding to the world's frozen schema/name catalog, and carry the agreed bootstrap receipt.

The similarly named CoreEntity Blueprint helpers are bounded v5 **header-metadata-only** documents. The ambiguous legacy nodes remain as deprecated wrappers and cannot create or load an executable journal. `FEAT-02` remains open for checkpoints, seeking, long-session streaming, and retention policy.

Design gates

Gate A - approved July 18, 2026

Owner-by-default deterministic authority with scoped grants; coordinator and match-administration capabilities remain separate from gameplay slots; exact versioned native/Blueprint handler registry; topology-neutral core with the Unreal relay as the first shipped transport adapter.

Gate B - open

Exact persistence contract for active Blueprint latent execution and opaque continuation fallback.

Gate C - open

Contextual authoritative-destination provider registry and allowed bypass flags.

Gate D - open

Cover scoring, contention, reservation lifecycle, queued orders, and moving providers.

Gate E - open

Height-aware FoW quality policy and evidence required to replace the default.

Gate F - open

Public API clusters: targeters, modifiers, terrain, production, team vision, and movement feel.

Gate G - approved July 22, 2026

Canonical match rules and exact peer receipt consensus; world-lifetime inert facade plus one self-culling transient transaction; immediate Blueprint deterministic-value contributions plus native registered contributors; GameMode remains the Unreal authority shell without duplicated deterministic defaults.

Related approved feature rows do not automatically close Gates C through E. Their full design questions remain open wherever the canonical ledger still labels them as gates.

Definition of complete

The campaign is complete only when every ledger row is:

- **Fixed** or **Verified**;
- **Disproved** with recorded evidence; or
- explicitly **Deferred** by RJ with a rationale.

The final acceptance matrix must include:

- [ ] Development, clean, and Shipping builds.
- [ ] Framework-only and supported extension combinations.
- [ ] Downstream public-consumer compilation.
- [ ] Canonical serial/parallel and fresh-process digest agreement.
- [ ] Snapshot restore plus next-N-tick equivalence.
- [ ] Replay, peer, and reconnect equivalence where applicable.
- [ ] Hostile-network and malformed-payload scenarios.
- [ ] Long-match memory and performance soak.
- [ ] Blueprint compile and content validation.
- [ ] Scripted PIE tactics-gym review and RJ acceptance for behavior-sensitive defaults.